

Sweep-Free Coherent Tracking Using a Multi-Tone Local Oscillator with a 100-MHz-Bandwidth Quadrant Photodetector

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Abstract *With narrowband detectors, coherent beam tracking is challenging under signal–local oscillator frequency offset uncertainty, given reliance on frequency sweeping or wideband reception. We demonstrate frequency-offset-tolerant error signal generation and closed-loop beam tracking under realistic disturbances using a multi-tone local oscillator. ©2026 The Author(s)*

Introduction

In space optical communication systems, accurate and stable beam tracking is essential to maintain a reliable communication link [1–5]. This requirement becomes particularly stringent for long-distance links such as Earth–Moon communications, where the received optical power is limited over large propagation distances, making high-sensitivity tracking a critical challenge. Coherent tracking using a quadrant photodetector (QPD) is a promising approach for achieving high sensitivity [6–10]. In this scheme, a local oscillator (LO) is employed at the receiver, and the signal and LO beams are optically mixed to generate a beat signal. Compared to direct detection, the coherent scheme can provide a sensitivity improvement of 10–20 dB under certain conditions [6].

A key challenge in coherent tracking is the precise control of the frequency offset between the signal and the LO. Since the tracking signal is derived from the signal–LO beat, this offset directly determines the beat frequency. Because the QPD cannot detect beat signals beyond its analogue bandwidth, the frequency offset must remain within this range, typically on the order of 100 MHz for practical QPDs. However, the absolute frequency uncertainty of practical laser sources is typically on the order of several gigahertz [12, 13], making it challenging to ensure that the signal–LO beat frequency initially lies within the QPD bandwidth.

A common approach to address this issue is to sweep the LO frequency across the uncertainty range until the beat is detected, which provides a straightforward and widely adopted means of frequency acquisition [7–9, 11]. Nevertheless, unlike direct-detection-based tracking, which can be initiated immediately, sweep-based approaches are inherently time-consuming due to their sequential nature. Coherent tracking is primarily motivated by its superior sensitivity and therefore becomes particularly advantageous in photon-limited scenarios, such as long-distance free-space links. Under these conditions, reliable beat detection requires sufficiently long integration times to achieve adequate signal-to-noise ratio (SNR). As a result, the allowable sweep rate is

constrained, causing the frequency acquisition time to become a dominant factor in the overall tracking initialization. An alternative strategy is to employ a wideband receiver front-end capable of detecting beat signals over the entire frequency uncertainty range, thereby eliminating the need for frequency sweeping [10]. While such wideband approaches enable rapid acquisition, they inherently rely on GHz-class analogue or digital signal processing, resulting in increased noise bandwidth and implementation complexity that can complicate noise management in practical receiver implementations.

In this work, we propose a sweep-free coherent QPD-based beam-tracking scheme using a multi-tone LO. By simultaneously covering the signal–LO frequency offset uncertainty with multiple optical tones, the proposed scheme enables immediate and frequency-offset-tolerant tracking with a narrowband receiver front-end, without requiring explicit LO frequency acquisition. While multi-tone LOs have been explored in coherent communication systems, this work demonstrates their application to beam tracking through experimental validation of stable error-signal generation and closed-loop tracking under realistic angular disturbances.

Experimental Setup and Principle

Figure 1 shows the experimental setup used to demonstrate the proposed approach. The key principle is that multiple discrete optical frequency components are generated simultaneously at the LO, instead of sequentially sweeping the LO frequency. These LO tones are designed to span the expected signal–LO frequency uncertainty such that at least one signal–LO beat component always falls within the analogue bandwidth of the QPD. As a result, tracking error signals can be generated immediately without any frequency sweeping or acquisition process, while maintaining a narrowband receiver front-end.

In the experiment, the beam to be tracked is referred to as the signal beam and is generated by a wavelength-tuneable laser operating at 1550 nm. Dynamic angular disturbances, emulating

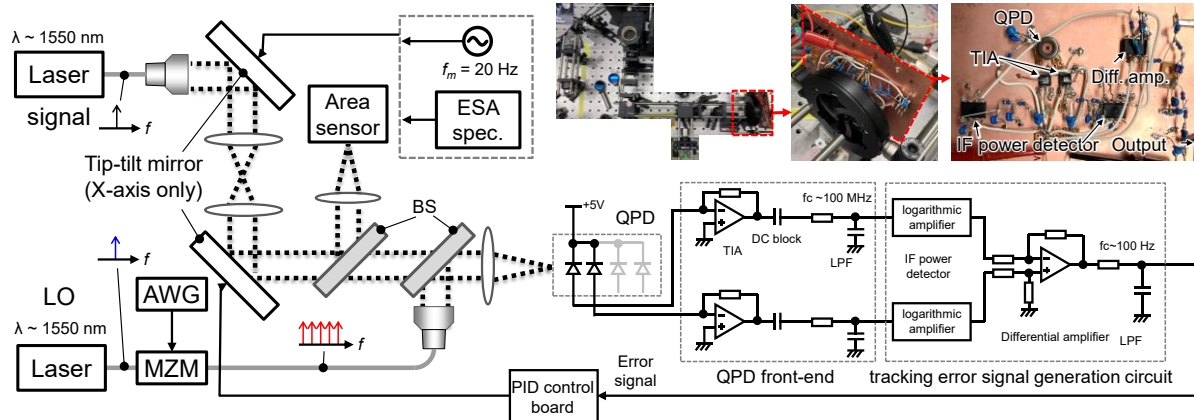


Fig. 1: Experimental setup for sweep-free coherent beam tracking using a multi-tone LO and a narrowband QPD.

satellite attitude jitter, are applied to the signal beam using a tip-tilt mirror and closed-loop beam tracking is performed using an additional tip-tilt mirror. For simplicity, the present experiment implements closed-loop tracking along a single transverse axis (x-axis). After propagation through the optical setup, the signal beam is combined with the LO and detected by a QPD. A commercially available QPD with an analogue bandwidth of approximately 100 MHz (Edmund Optics, #17-077 [14]) is used to generate tracking error signals. The LO is generated by another independent wavelength-tuneable laser and input to a lithium-niobate Mach-Zehnder modulator (MZM) driven by a radio-frequency multi-tone electrical signal synthesized by an arbitrary waveform generator (AWG). As a result, multiple optical sidebands are generated on the LO, forming a multi-tone optical spectrum. Although only single-axis tracking is demonstrated here for simplicity, extension to two-dimensional tracking is straightforward.

The photocurrents from the QPD are processed by an analogue front-end with a transimpedance amplifier (TIA) and a low-pass filter, followed by logarithmic intermediate frequency (IF) power detection to generate an analogue

tracking error signal. This error signal is fed back to a tip-tilt mirror through an analogue PID control board, forming a fully analogue closed-loop beam tracking system, as shown in Fig. 1. The signal processing bandwidth is limited to approximately 100 MHz, allowing the entire receiver and control circuitry to be implemented using low-frequency discrete electronic components. A photograph of the implemented analogue circuit is shown in the inset of Fig. 1.

In the present experiment, a total signal-LO frequency uncertainty of ± 5 GHz is assumed, combining a ± 1.5 GHz absolute frequency uncertainty from the CCSDS standard [13] and a Doppler shift of up to approximately ± 2 GHz in lunar-GEO optical links [15]. Given the QPD analogue bandwidth of approximately 100 MHz, the tone spacing is set to 200 MHz, and multiple optical tones are generated around the LO centre frequency to cover the uncertainty range. Figure 2 shows the measured optical spectrum of the generated multi-tone LO, confirming that multiple discrete optical tones are simultaneously generated over the designed frequency range.

Experimental Demonstration

We first experimentally verify that the proposed multi-tone LO enables sweep-free generation of QPD tracking error signals under large signal-LO frequency offsets. In this measurement, the LO frequency was fixed while the signal laser frequency was tuned. A 20 Hz sinusoidal angular disturbance with constant amplitude was applied to the signal beam. Figure 3 compares the normalised tracking error signal power obtained using the proposed multi-tone LO and a conventional single-tone LO. With the single-tone LO, the error signal power rapidly degrades when the signal-LO frequency offset exceeds approximately 100 MHz. In contrast, with the multi-tone LO, continuous tracking error signals are obtained over a wide frequency offset range from -5.4 GHz to $+5.5$ GHz. These results demonstrate that the proposed approach enables

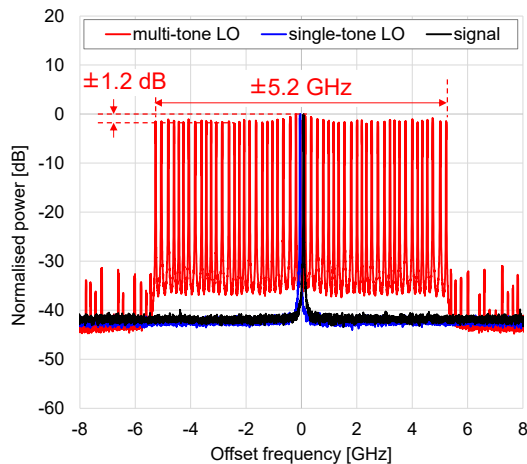


Fig. 2: Optical spectra of the signal and the multi-tone LO.

sweep-free error signal generation even under multi-gigahertz signal–LO frequency offset uncertainty.

Based on the above results, we next demonstrate closed-loop beam tracking using the proposed coherent tracking architecture. To evaluate robustness under realistic disturbance conditions, angular disturbances based on ESA specifications [16,17] are applied. Figure 4 shows time-domain waveforms of the angular deviation measured by an area sensor, with the tracking loop switched off and on. The angle fluctuation is largely suppressed when tracking is enabled, and the root-mean-square (RMS) angular deviation is reduced to approximately 33% of that without tracking. Figure 5 shows the power spectral density (PSD) of the angular deviation corresponding to the time-domain data in Fig. 4. The results indicate substantial suppression of angular disturbances, particularly in the low-frequency region, which significantly contributes to compensating the pointing errors under realistic satellite attitude disturbances. These results confirm effective disturbance suppression through closed-loop beam tracking while preserving a narrowband receiver front-end.

Discussion

In multi-tone coherent operation, the available LO power is distributed among multiple optical tones, resulting in reduced per-tone signal levels. However, from a fundamental perspective, direct detection yields a signal proportional to the received signal power $P_s (= \sqrt{P_s P_s})$, whereas coherent detection produces a beat term proportional to $\sqrt{P_s P_{LO}}$ [18]. In this sense, even when the LO power is divided among multiple tones, the favorable signal scaling of coherent detection can be partially preserved in the regime $P_{LO} \gg P_s$, which is particularly relevant for weak-signal scenarios such as long-distance links, with the practically achievable performance determined by implementation-specific SNR.

An important aspect of the proposed approach, compared with sweep-based coherent scheme, is that the handling of signal–LO frequency offset uncertainty is shifted from the receiver front-end to the LO generation stage. Since the receiver processes weak optical signals, maintaining a narrow analogue bandwidth is beneficial for achieving low-noise operation. In contrast, multi-tone LO generation can be implemented under higher signal power conditions, which relaxes noise constraints and makes GHz-class operation more practical using existing components such as electro-optic modulators. This separation enables frequency-offset-tolerant coherent beam tracking while preserving a low-noise, narrowband receiver architecture, and

eliminates the potentially long initialization time associated with frequency sweeping in single-tone coherent tracking.

Conclusion

We have demonstrated a sweep-free coherent beam tracking scheme using a multi-tone LO and a 100-MHz-bandwidth QPD. The results confirm frequency-offset-tolerant error signal generation and effective closed-loop beam tracking under realistic disturbance conditions with a narrowband receiver front-end.

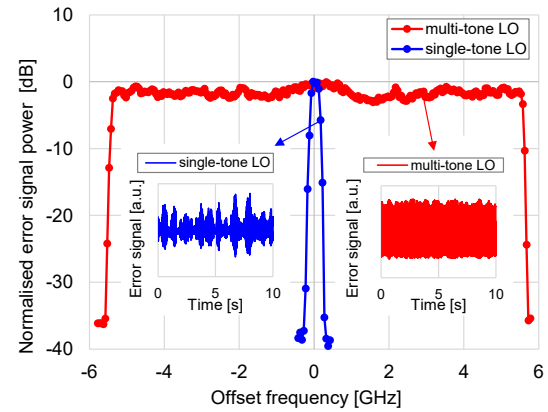


Fig. 3: Normalised QPD tracking error signal power versus signal–LO frequency offset.

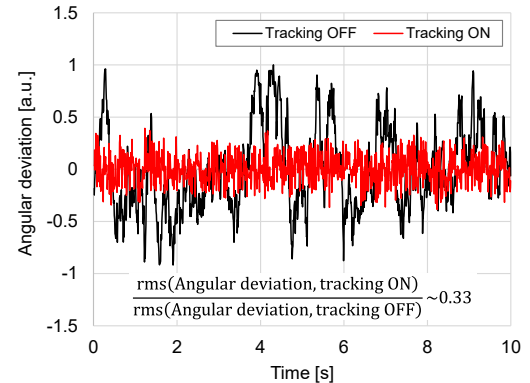


Fig. 4: Closed-loop beam tracking using the proposed multi-tone coherent scheme.

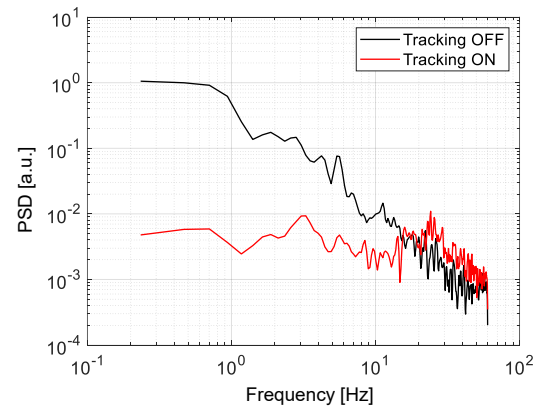


Fig. 5: Power spectral density of tracking error under ESA spec.-based angular disturbances.

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